

Problem Statement:

It is undeniable that our country relies on the work of farmers. As a matter of fact, the liveliness of these farmers depends on debt financing. Increasingly, American farmers are unable to pay back the loans they took out to produce their harvest. According to the USDA, farm loan delinquency rates have been climbing steadily, and farm bankruptcies have risen sharply in recent years. Beyond just the individual farmer, we also worry that this will create a trickle-down effect that will adversely impact all lowans through disruptions in the food chain.

Relevance to Iowa:

As it turns out, the stereotype of Iowa as an agriculture powerhouse is true; according to the Iowa Farm Bureau, our state has 86,911 farms that cover tens of millions of acres. Despite our seismic production, Iowa farmers carry some of the heaviest debt loads in the country: over \$1.8 million for large farms and roughly \$239,500 for small farms, according to the Economic Research Service. Farms failing create a ripple effect in which banks tighten lending, supply chains are disrupted, small towns lose business, and more. In order to assuage these adverse outcomes, a tool that advises farmers where to plant crops specifically would cut wasteful spending and increase farmer profits.

Proposed Solution:

Our solution involves creating a multi-modal AI model that takes soil inputs --- pictures, pH readings, and other data --- and determines soil quality and subsequently creates a "hotspot" map of soil quality (good, medium, poor). As such, farmers can use this tool to purchase materials accordingly and only harvest in areas in which the soil quality is good (or improve the soil quality of deficient areas). Consequently, farmers will allocate resources more efficiently and profits will increase, overall ameliorating an otherwise grim industry that serves as the backbone of our nation.

Feasibility Analysis:

Our proposed soil-quality hotspot mapping solution is achievable within the given timeline because we will use existing datasets and pretrained models rather than collecting new data or building from scratch. The core inputs we need are publicly accessible. Soil composition data, such as pH, organic matter, and nutrient levels, are available from USDA and state-level agricultural databases. Satellite imagery and vegetation indices are publicly available through platforms like Landsat and Sentinel.

Farmers may also provide localized soil test data or smartphone or drone images. Because our model integrates both structured (pH readings, moisture levels, nutrient content) and unstructured (soil images) data, it is multimodal and relies on realistic, obtainable inputs.

In this project, we will use a staged modeling pipeline. For image processing, we will use a pretrained convolutional neural network (CNN) to extract features from soil images. For a tabular data model, we will analyze soil test readings and environmental variables using supervised learning. Finally, for our fusion layer, we will combine outputs from both modalities to produce a final soil-quality classification. This modular approach will ensure that development remains manageable.

The project can realistically be completed in phases aligned with the competition milestones:

- **Phase 1–2:** Data sourcing, cleaning, and exploratory analysis
- **Phase 3:** Train individual models for image and tabular data
- **Phase 4:** Integrate modalities and build a hotspot mapping prototype
- **Phase 5:** Evaluate model accuracy and refine usability

Because we are using existing tools such as PyTorch and publicly available datasets, model development can begin early without waiting on proprietary data.

Some potential problems include variability in data quality. Soil conditions fluctuate across regions and seasons, potentially leading to noise. To address this, we can use normalization techniques and region-specific modeling where appropriate. Also, soil recommendations may not transfer equally across crop types or farm sizes. We will address this by incorporating crop-type metadata as an additional feature in our model.

Overall, this solution is achievable within the challenge timeline. By focusing on a practical prototype rather than a fully commercialized system, our team can deliver a functional tool that promptly addresses a real economic challenge facing Iowa farmers.